Faculty of Engineering

Topics in Computer Science Scientific Research

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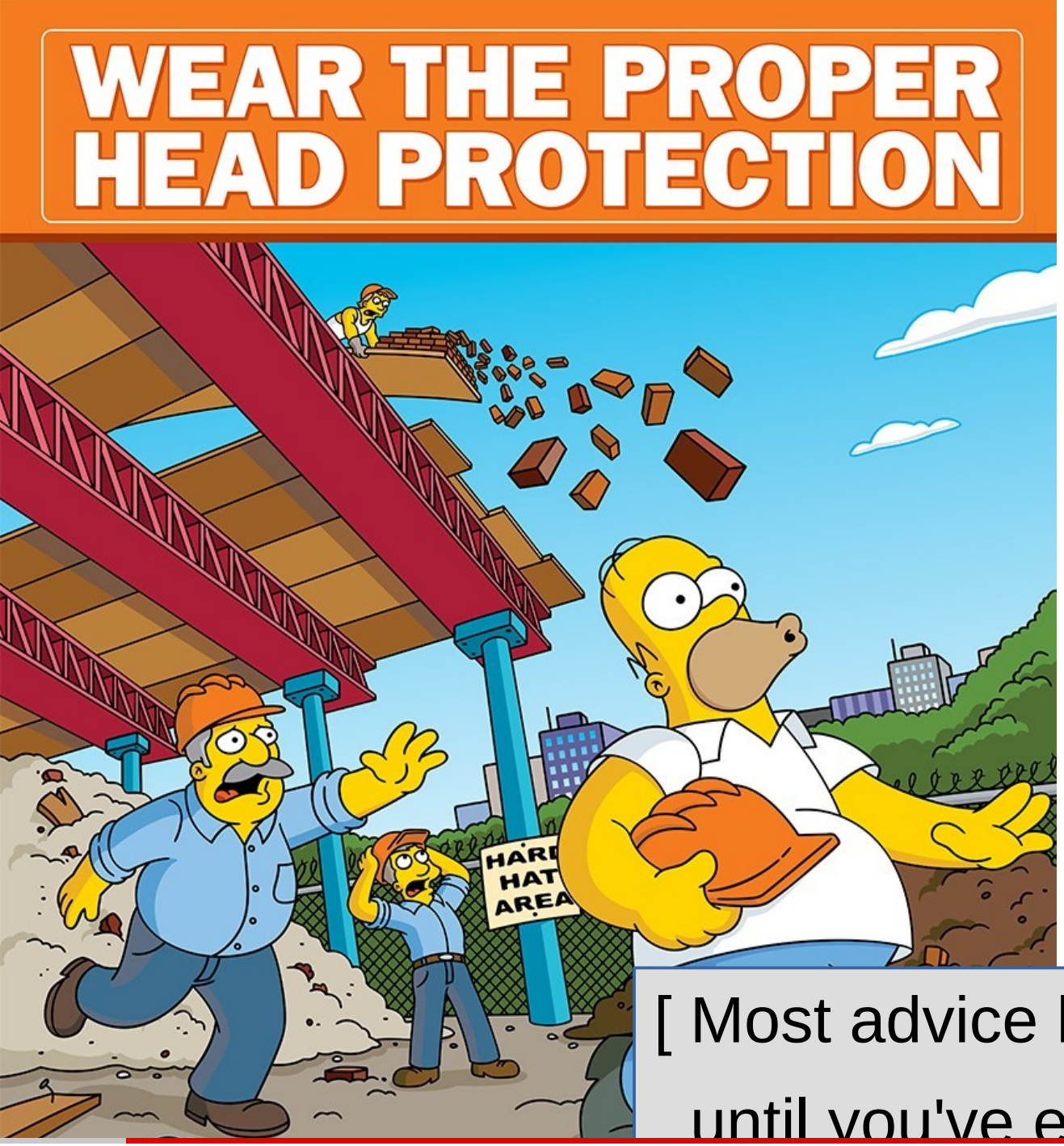
de ciencias Aplicadas

Sources

- How to do good research, get it published in SIGKDD and get it cited
- Tips for a succesful PhD, and how to win an award with it!



[Most advice has little meaning, until you've experienced it]

Pieces

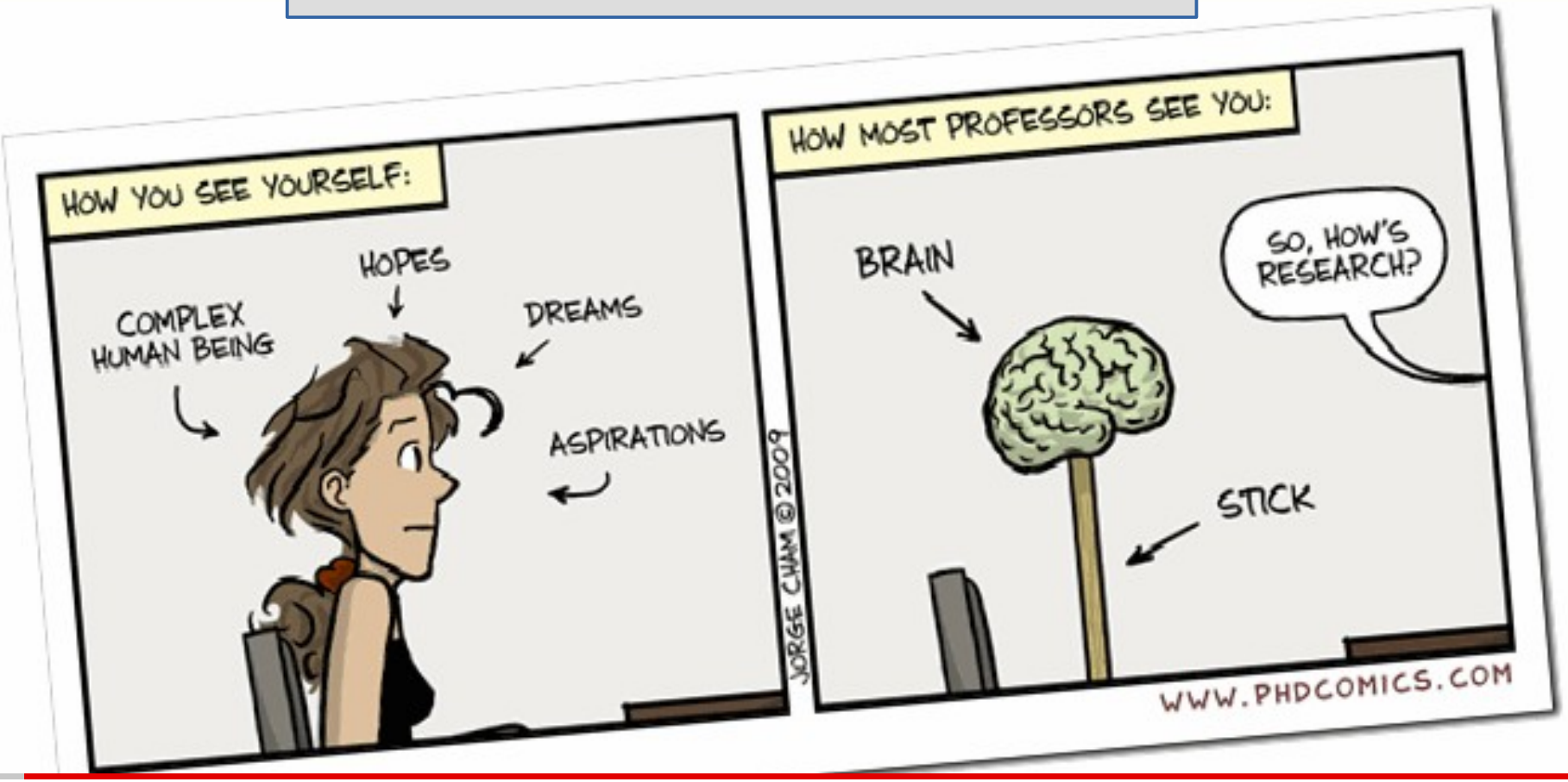
- Working with your promotor
- Writing a paper
- Conference participation
- Writing an (award winning) PhD
- After the PhD

Working with your promotor

[You will feel stupid]

Working with your promotor

[Guard your own interests]



Working with your promotor

[Embrace feedback]

Working with your promotor

[Learn how to disagree]

Writing a paper

[Tell a story]

Writing a paper

common in spreadsheets, with semantically related items stored in rows or columns and formula's usually ranging over contiguous ranges, as formulas parallel to a table are often *dragged* over its entire span. Our contributions are as follows:

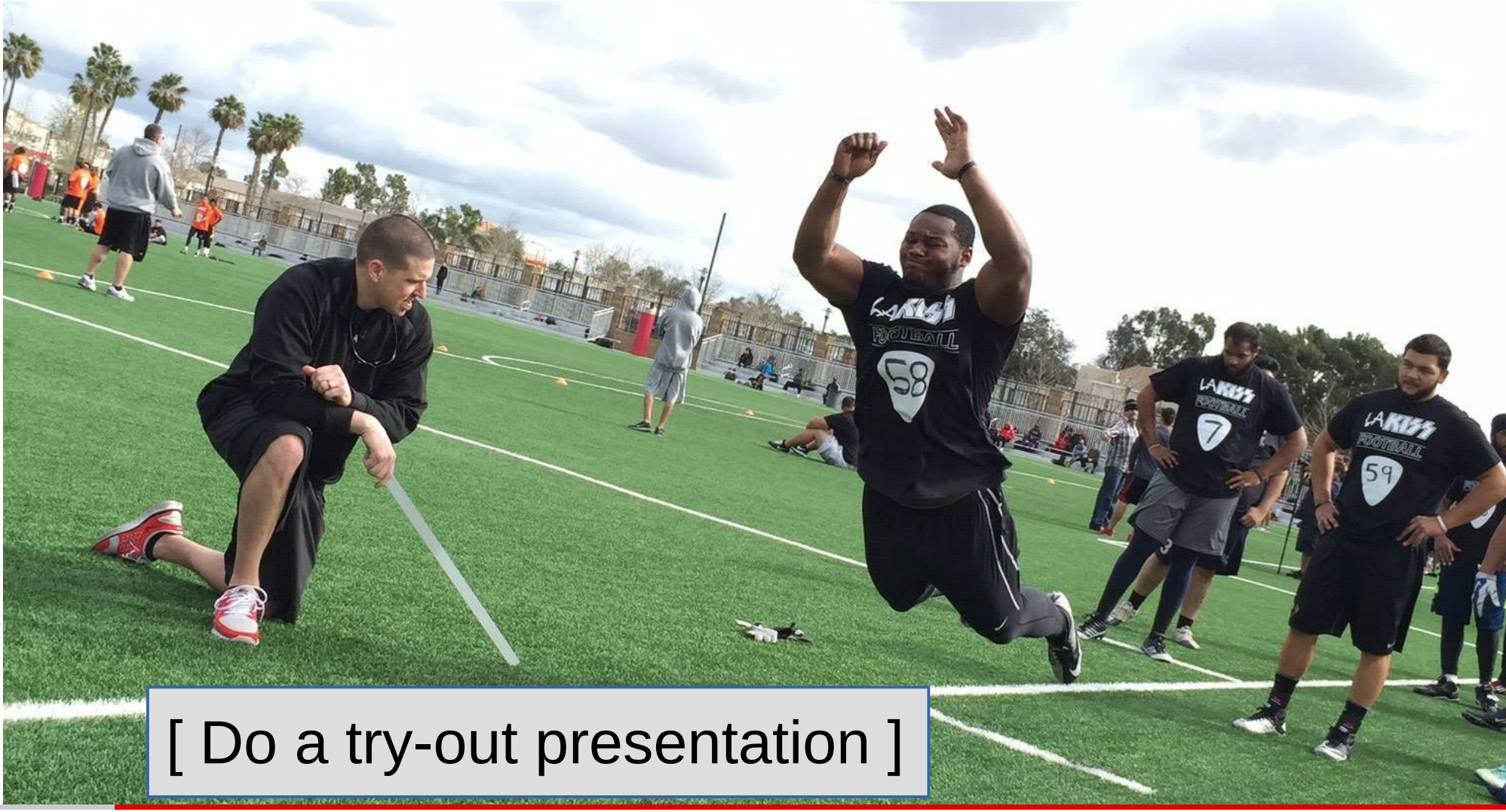
- we define the tabular constraint learning problem, where the goal is to find constraints that range over entire rows or columns in an unsupervised way;
- we propose an effective two-stage generate-and-test method where the first stage reasons only over properties of contiguous blocks of rows/columns, and the second stage continues to investigate individual rows and columns and their content;
- furthermore, in the first stage we use a constraint solver to efficiently enumerate all combinations of maximally contiguous blocks compatible with the arguments of the candidate constraints
- experiments on different publicly available spreadsheets show that the system is able to extract constraints with high precision and recall.

This paper is organized as follows. Section 2 discusses related work. Section 3 introduces the relevant concepts and defines the problem formally while Sect. 4 presents our approach. This is evaluated in Sect. 5 after which we show how to realize two of the motivating applications using our system in Sect. 6, and we conclude in Sect. 7.

Writing a paper

[Align section content, iteratively]

Conference participation



[Do a try-out presentation]

Conference participation

[Talk to other people]

speaker: “Hi, I saw your presentation and liked ...”
non-prof: “Hi, are you presenting at the conference?”
prof: “Hi, I've been reading a bit about your work on … and like the direction of ...”

Community participation

[Have a website]

Papers, software, data

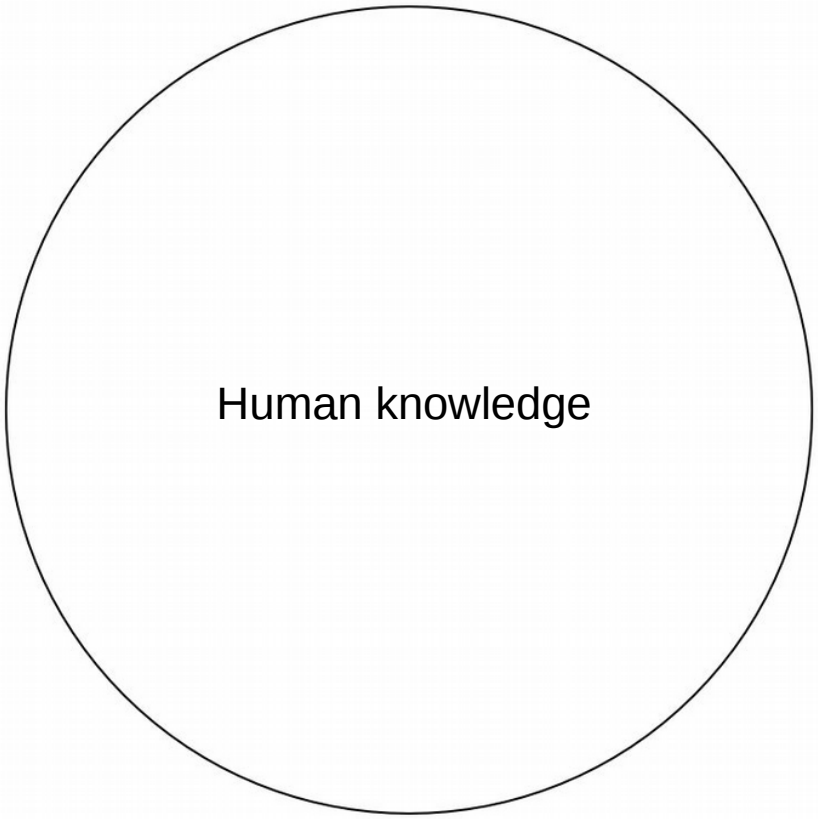
Github...

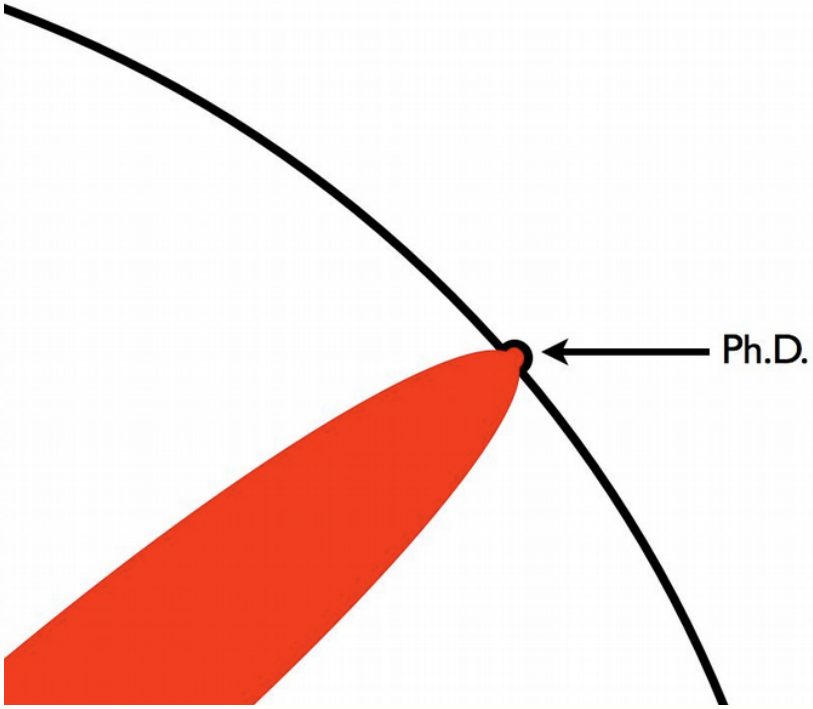
Twitter, research gate?

Writing an (award winning) PhD

[Focus on intro & conclusion, iteratively]

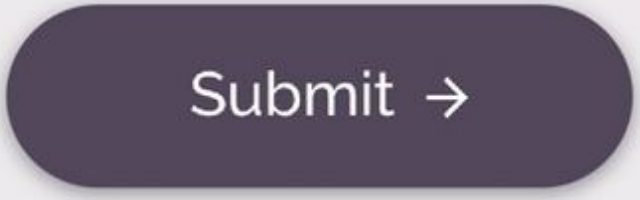
Writing an (award winning) PhD





[Write about your impact]

Writing an (award winning) PhD



[You only win if you apply]

After the PhD

[PostDoc @ jury member?]

After the PhD

[PostDoc? Industry?

Ask and mail anybody]

After the PhD

Outline

- The Review Process
- Writing a SIGKDD paper
  - Finding problems/data
    - Framing problems
    - Solving problems
  - Tips for writing
    - Motivating your work
    - Clear writing
    - Clear figures
- The top ten reasons papers get rejected
  - With solutions

The Curious Case of Srikanth Krishnamurthy

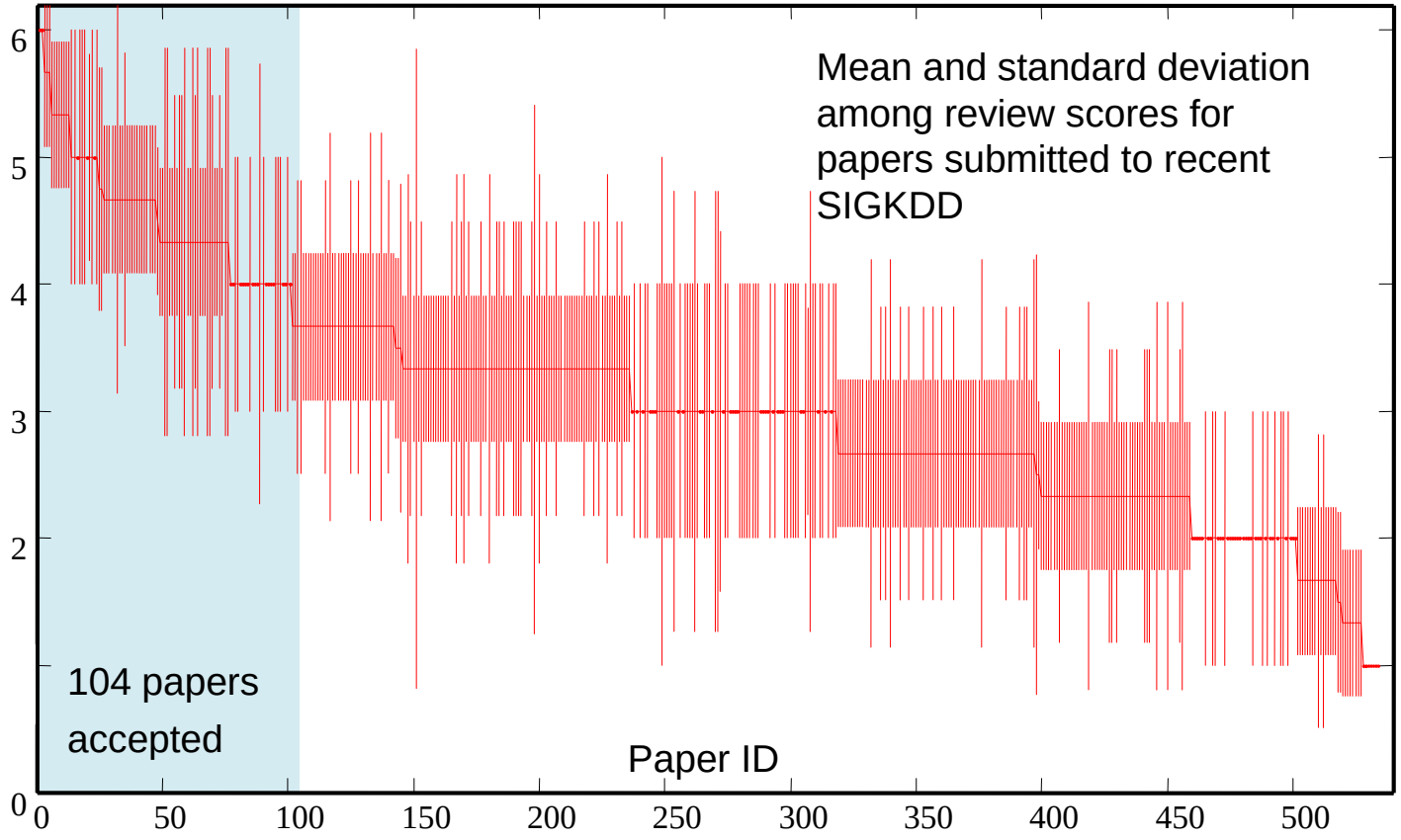
- In 2004 Srikanth's student submitted a paper to MobiCom
- Deciding to change the title, the student resubmitted the paper, accidentally submitting it as a new paper
- One version of the paper scored 1,2 and 3, and was rejected, the other version scored a 3,4 and 5, and was accepted!

- This "natural" experiments suggests that the reviewing process is random, is it really that bad?

A look at the reviewing statistics for a recent SIGKDD

(I cannot say what year)

Mean number of reviews 3.02

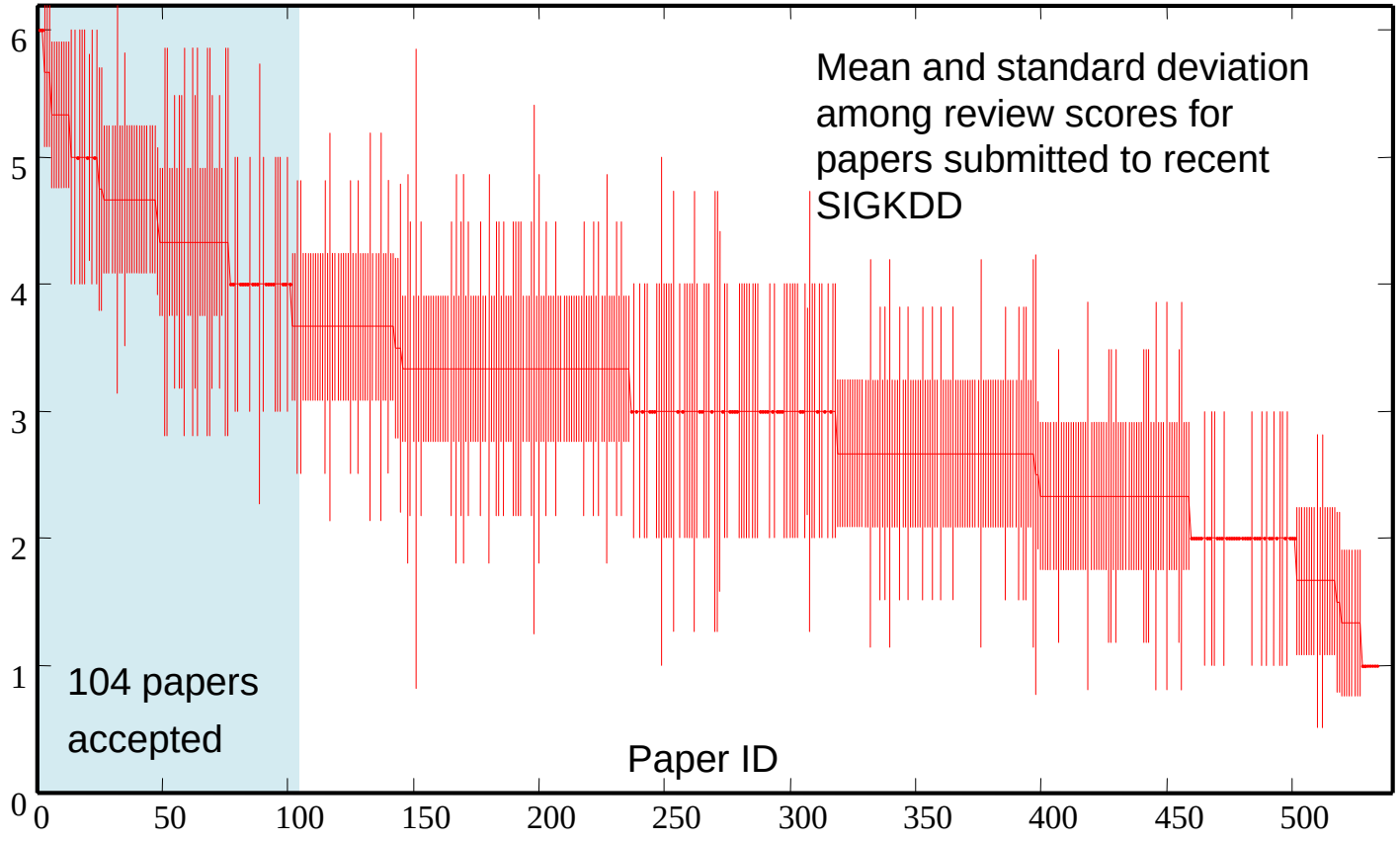


- Papers accepted after a discussion, not solely based on the mean score.
- These are final scores, after reviewer discussions.
- The variance in reviewer scores is **much larger** than the differences in the mean score, for papers on the boundary between accept and reject.
- In order to *halve* the standard deviation we must *quadruple* the number of reviews.

Conference reviewing is an imperfect system.

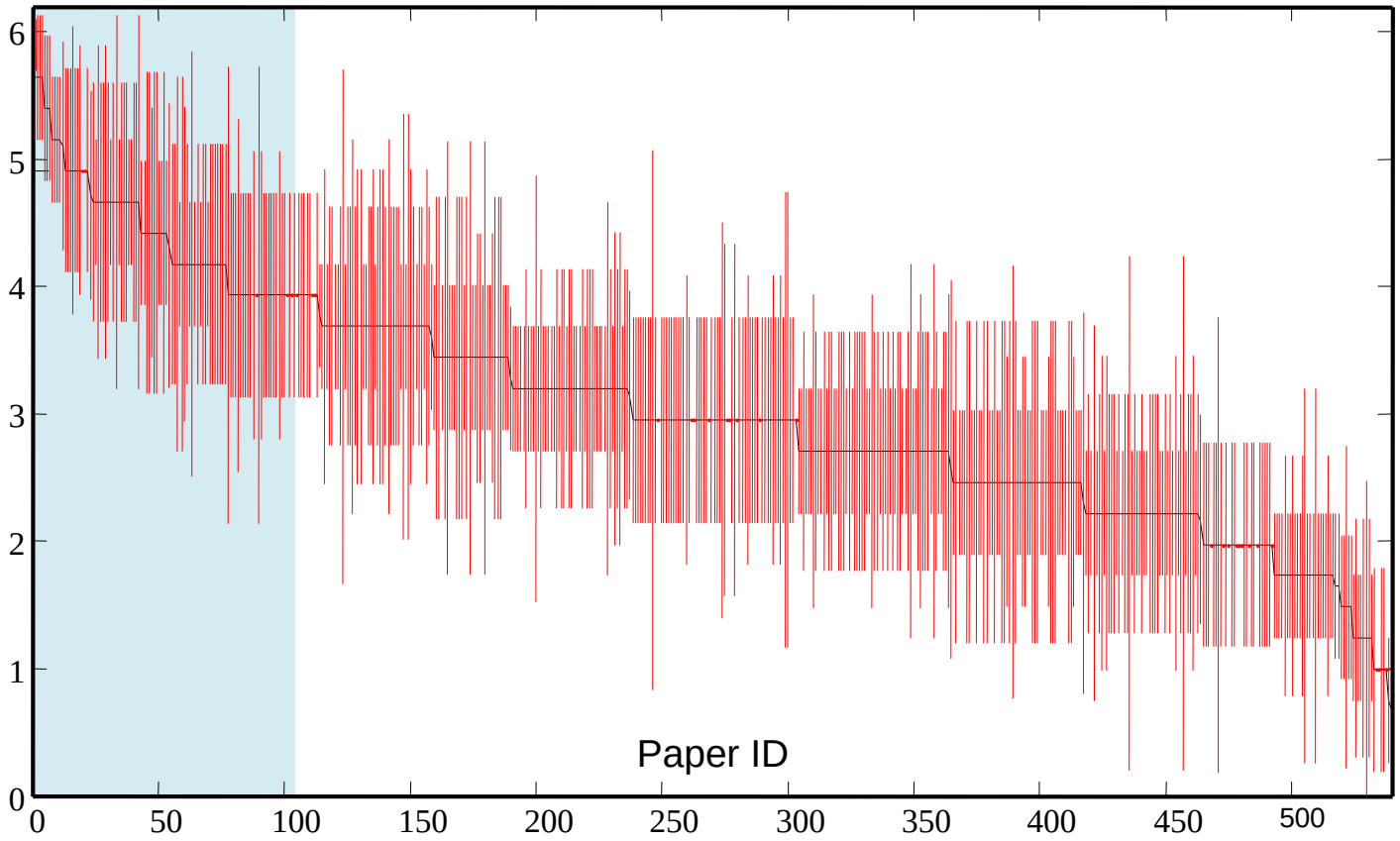
We must learn to live with rejection.

All we can do is try to make sure that our paper lands as far left as possible



- At least three papers with a score of 3.67 (or lower) must have been accepted. But there were a total of 41 papers that had a score of 3.67.
- That means there exist at least 38 papers that were rejected, that had the same or better numeric score as some papers that were accepted.
- **Bottom Line**: With very high probability, multiple papers will be rejected in favor of less worthy papers.

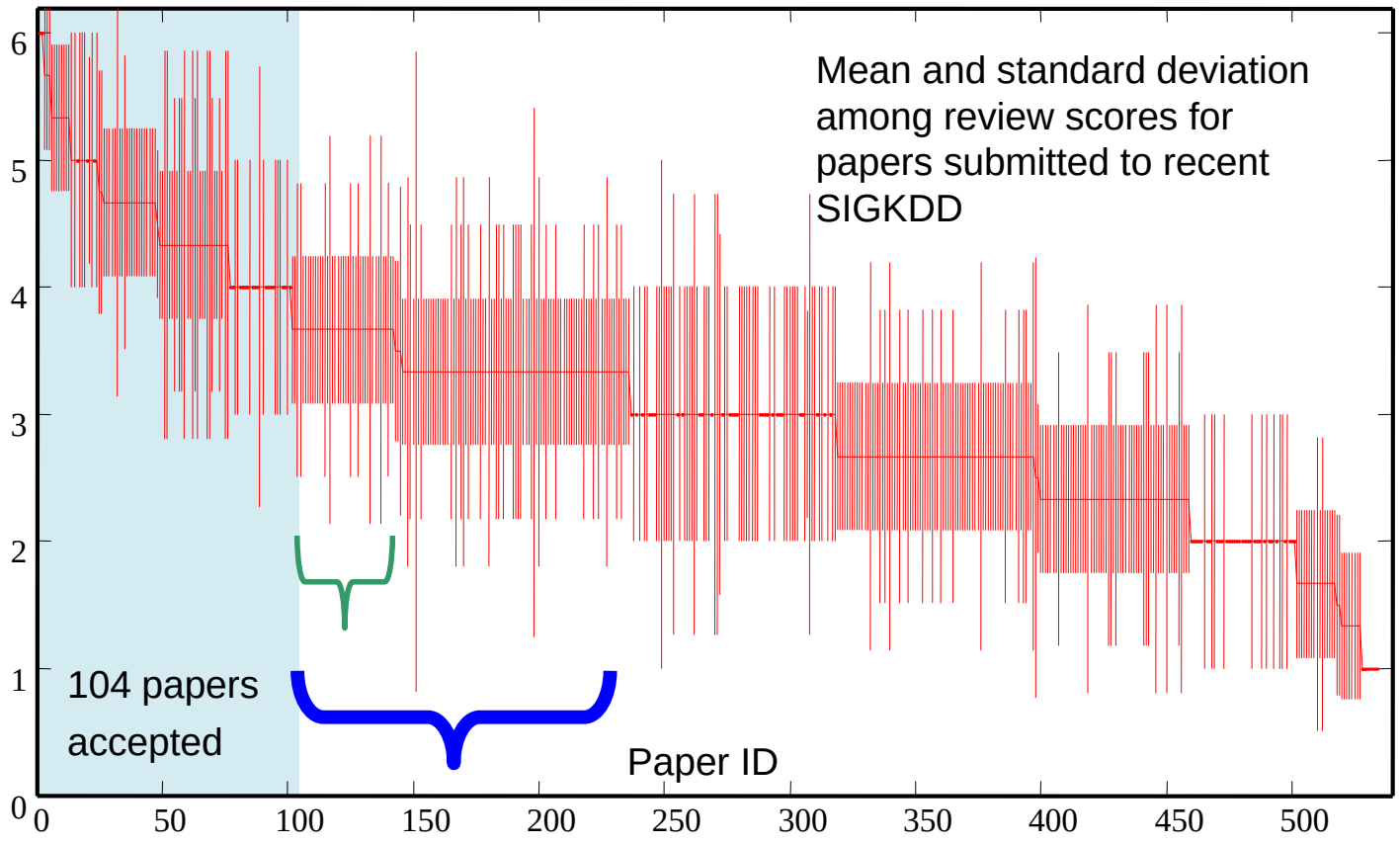
A sobering experiment



- Suppose I add one *reasonable review* to each paper.
- A *reasonable review* is one that is drawn uniformly from the range of one less than the lowest score to one higher than the highest score.
- If we do this, then on average, 14.1 papers move across the accept/reject borderline. This suggests a very brittle system.

But the good news is…

Most of us only need to improve a little to improve our odds a lot.



- Suppose you are one of the 41 groups in the **green** (light) area. If you can convince just one reviewer to increase their ranking by just one point, you go from near certain reject to near certain accept.
- Suppose you are one of the 140 groups in the **blue** (**bold**) area. If you can convince just one reviewer to increase their ranking by just one point, you go from near certain reject to a good chance at accept.

Idealized Algorithm for Writing a Paper

- Find problem/data
- Start writing (*yes*, start writing *before* and *during* research)
- Do research/solve problem
- Finish 95% draft
- Send preview to mock reviewers
- Send preview to the rival authors (virtually or literally)
- Revise using checklist.
- Submit

One month before deadline

What Makes a Good Research Problem?

- **It is important:** If you can solve it, you can make money, or save lives, or help children learn a new language, or...
- **You can get real data**: Doing DNA analysis of the Loch Ness Monster would be interesting, but…
- **You can make incremental progress**: Some problems are all-or-nothing. Such problems may be too risky for young scientists.
- **There is a clear metric for success**: Some problems fulfill the criteria above, but it is hard to know when you are making progress on them.

Finding Problems/Finding Data

- Finding a good problem can be the hardest part of the whole process.
- Once you have a problem, you will need data…
- As I shall show in the next few slides, finding problems and finding data are best integrated.
- However, the obvious way to find problems is the best, read *lots* of papers, both in SIGKDD and elsewhere.

Domain Experts as a Source of Problems

- Data miners are almost unique in that they can work with almost any scientist or business
- I have worked with anthropologists, nematologists, archaeologists, astronomers, entomologists, cardiologists, herpetologists, electroencephalographers, geneticists, space vehicle technicians etc
- Such collaborations can be a rich source of interesting problems.

Working with Domain Experts I

- Getting problems from domain experts might come with some bonuses
- Domain experts can help with the **motivation** for the paper
  - *..insects cause 40 billion dollars of damage to crops each year..*
  - *..compiling a dictionary of such patterns would help doctors diagnosis..*
  - *Petroglyphs are one of the earliest expressions of abstract thinking, and a true hallmark...*
- Domain experts sometimes have funding/internships etc
- Co-authoring with domain experts can give you credibility.

Augmenting the Generalized Hough Transform to Enable the Mining of Petroglyphs

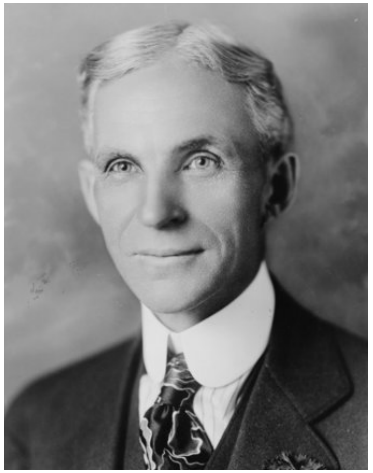
Qiang Zhu Xiaoyue Wang Eamonn Keogh [1]Sang-Hee Lee

Dept. of Computer Science & Engineering, [1]Dept. of Anthropology

University of California, Riverside, CA 92521

Working with Domain Experts II

If I had asked my customers what they wanted, they would have said a faster horse



Henry Ford

- Ford focused not on stated need but on **latent need**.
- In working with domain experts, don't just ask them what they want. Instead, try to learn enough about their domain to understand their **latent** needs.
- In general, domain experts have little idea about what is hard/easy for computer scientists.

Working with Domain Experts III

Concrete Example:

- I once had a biologist spend an hour asking me about sampling/estimation. She wanted to estimate a quantity.
- After an hour I realized that we did not have to estimate it, we could compute an *exact* answer!
- The exact computation did take three days, but it had taken several years to gather the data.
- Understand the **latent need**.

Finding Research Problems

- Suppose you think idea **X** is very good
- Can you extend **X** by…
  - Making it more accurate (*statistically significantly* more accurate)
  - Making it faster (usually an order of magnitude, or no one cares)
  - Making it an anytime algorithm
  - Making it an online (streaming) algorithm
  - Making it work for a different data type (including uncertain data)
  - Making it work on low powered devices
  - Explaining *why* it works so well
  - Making it work for distributed systems
  - Applying it in a novel setting (industrial/government track)
  - Removing a parameter/assumption
  - Making it disk-aware (if it is currently a main memory algorithm)

Finding Research Problems (examples)

- Suppose you think idea X is a very good
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- The Nearest Neighbor Algorithm is very useful. I wondered if we could make it an ***anytime*** algorithm…. ICDM06 [b].
- Motif discovery is very useful for ***DNA***, would it be useful for ***time series***? SIGKDD03 [c]
- The bottom-up algorithm is very useful for batch data, could we make it work in an ***online*** setting? ICDM01 [d]
- Chaos Game Visualization of DNA is very useful, would it be useful for ***other kinds of data***? SDM05 [a]

[a] Kumar, N., Lolla N., Keogh, E., Lonardi, S. , Ratanamahatana, C. A. and Wei, L. (2005). Time-series Bitmaps: ICDM 2006

Finding Research Problems

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- Some people have suggested that this method can lead to incremental, boring, low-risk papers…
  - Perhaps, but there are 104 papers in SIGKDD this year, they are not all going to be groundbreaking.
  - Sometimes ideas that seem incremental at first blush may turn out to be very exciting as you explore the problem.
  - An early career person might eventually go on to do high risk research, after they have a "cushion" of two or three lower-risk SIGKDD papers

Framing Research Problems I

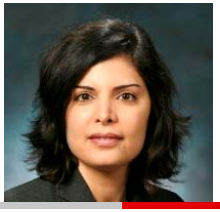
As a reviewer, I am often frustrated by how many people don't have a clear problem statement in the abstract (or the entire paper!)

Can you write a research statement for your paper in a single sentence?

- **X** is good for **Y** (in the context of **Z**).
- **X** can be extended to achieve **Y** (in the context of **Z**).
- The adoption of **X** facilitates **Y** (for data in **Z** format).
- An **X** approach to the problem of **Y** mitigates the need for **Z**.

(An **anytime algorithm** approach to the problem of **nearest neighbor classification** mitigates the need for **high performance hardware**) (Ueno et al. ICDM 06)

If I, as a reviewer, cannot form such a sentence for your paper after reading just the abstract, then your paper is usually doomed.



I hate it when a paper under review does not give a concise definition of the problem

Framing Research Problems II

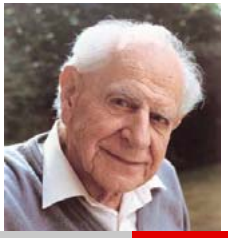
Your research statement should be **falsifiable**

A real paper claims:

> *To the best of our knowledge, this is most sophisticated subsequence matching solution mentioned in the literature.*

Is there a way that we could show this is not true?

Falsifiability (or **refutability**) is the logical possibility that an claim can be shown false by an observation or a physical experiment. That something is 'falsifiable' does not mean it is false; rather, that *if* it is false, then this can be shown by observation or experiment



Falsifiability is the demarcation between

Framing Research Problems III

Examples of falsifiable claims:

- *Quicksort is faster than bubblesort.* (this may needed expanding, *if the lists are..*)
- *The X function lower bounds the DTW distance.*
- *The L2 distance measure generally outperforms L1 measure*

  (this needs some work (under what conditions etc), but it is falsifiable)

Examples of unfalsifiable claims:

- *We can approximately cluster DNA with DFT.*
  - Any random arrangement of DNA could be considered a "clustering".
- *We present an alterative approach through Fourier harmonic projections to enhance the visualization. The experimental results demonstrate significant improvement of the visualizations.*
  - Since "*enhance*" and "*improvement* " are subjective and vague, this is unfalsifiable. Note that it *could* be made falsifiable. Consider:
    - *We improve the mean time to find an embedded pattern by a factor of ten.*

From the Problem to the Data

- At this point we have a concrete, falsifiable research problem
- Now is the time to get data!

  By "now", I mean months before the deadline. I have one of the largest collections of free datasets in the world. Each year I am amazed at how many emails I get a few days before the SIGKDD deadline that asks "*we want to submit a paper to SIGKDD, do you have any datasets that..*"
- Interesting, real (large, when appropriate) datasets *greatly* increase your papers chances.
- Having good data will also help do better research, by preventing you from converging on unrealistic solutions.
- Early experience with real data can feed back into the *finding and framing the research question* stage.

- Given the above, we are going to spend some time considering data..

Is it OK to Make Data?

There is a **huge** difference between…

We wrote a Matlab script to create random trajectories

and…

We glued tiny radio transmitters to the backs of Mormon crickets and tracked the trajectories

Why is Synthetic Data so Bad?

Suppose you say "*Here are the results on our synthetic dataset*:"

	Our Method	Their Method
Accuracy	95%	80%

This is good right? After all, you are doing much better than your rival.

Why is Synthetic Data so Bad?

Suppose you say "*Here are the results on our synthetic dataset*:"

	Our Method	Their Method
Accuracy	**95%**	**80%**

But as far as I know, you might have created ten versions of your dataset, but only reported one!

Even if you did not do this consciously, you may have done it unconsciously.

At best, *your* making of *your* test data is a huge conflict of interest

	Our Method	Their Method
Accuracy	80%	85%
Accuracy	75%	85%
Accuracy	90%	90%
Accuracy	**95%**	**80%**

Why is Synthetic Data so Bad?

Note that is does not really make a difference if you have real data but you modify it somehow, it is still synthetic data.

A paper has a section heading: **Results on Two Real Data Sets**

But then we read…

> *We add some noises to a small number of shapes in both data sets to manually create some anomalies.*

Is this still real data? The answer is *no*, even if they authors had explained how they added noise (which they don't).

Note that there are probably a handful of circumstances were taking real data, doing an

Synthetic Data can lead to a Contradiction

- Avoid the contradiction of claiming that the problem is very important, but there is no real data.
- If the problem is as important as you claim, a reviewer would wonder why there is no real data.
- I encounter this contradiction very frequently, here is a real example:

> - **Early in the paper**: *The ability to process large datasets becomes more and more important…*
> - **Later in the paper**: *..because of the lack of publicly available large datasets…*

How big does my Dataset need to be?

It depends…

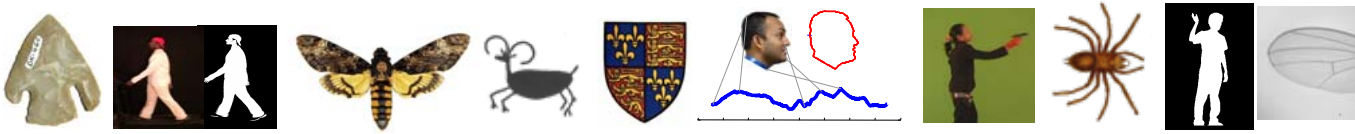
Suppose you are proposing an algorithm for mining Neanderthal bones. There are only a few hundred specimens known, and it is very unlikely that number will double in our lifetime. So you could reasonably test on a synthetic* dataset with a mere 1,000 objects.

However…

Suppose you are proposing an algorithm for mining Portuguese web pages (there are billions) or some new biometric (there may soon be millions). You do have an obligation to test on large datasets.

It is increasing difficult to excuse data mining papers testing on small datasets. Data is typically free, CPU cycles are essentially free, a terabyte of storage costs less than $100…

Where do I get Good Data?

- From your domain expert collaborators:
- From formal data mining archives:
  - The UCI Knowledge Discovery in Databases Archive.
  - The UCR Time Series and Shape Archive.
- From general archives:
  - Chart-O-Matic
  - NASA GES DISC
- From creating it: 
  - Glue tiny radio transmitters to the backs of Mormon crickets…
  - By a Wii, and hire a ASL interpreter to…
- Remember there is *no* excuse for not getting real data.

Solving Problems

- Now we have a problem and data, *all* we need to do is to solve the problem.
- Techniques for solving problems depend on your skill set/background and the problem itself, however I will quickly suggest some simple general techniques.
- Before we see these techniques, let me suggest you **avoid complex solutions**. This is because complex solutions...
  - …are less likely to generalize to datasets.
  - …are much easer to overfit with.
  - …are harder to explain well.
  - …are difficult to reproduce by others.

Unjustified Complexity I

From a recent paper:

This forecasting model integrates a case based reasoning (CBR) technique, a Fuzzy Decision Tree (FDT), and Genetic Algorithms (GA) to construct a decision-making system based on historical data and technical indexes.

- Even if you believe the results. Did the improvement come from the CBR, the FDT, the GA, or from the combination of two things, or the combination of all three?
- In total, there are more than 15 parameters…
- How reproducible do you think this is?

Unjustified Complexity II

- There *may* be problems that really require very complex solutions, but they seem rare. see [a].
- Your paper is implicitly claiming "*this is the simplest way to get results this good*".
- Make that claim *explicit*, and carefully justify the complexity of your approach.

[a] R.C. Holte, *Very simple classification rules perform well on most commonly used datasets*, Machine Learning 11 (1) (1993). This paper shows that one-level decision trees do very well most of the time.

J. Shieh and E. Keogh *iSAX: Indexing and Mining Terabyte Sized Time Series*. SIGKDD 2008. This paper shows that the simple

Unjustified Complexity III

Charles Elkan

Paradoxically and wrongly, sometimes if the paper used an excessively complicated algorithm, it is more likely that it would be accepted

If your idea is simple, *don't* try to hid that fact with unnecessary padding (although unfortunately, that does seem to work *sometimes*). Instead, *sell* the simplicity.

> *"…it reinforces our claim that our methods are very **simple** to implement.. ..Before explaining our **simple** solution this problem……we can objectively discover the anomaly using the **simple** algorithm…"* SIGKDD04

Simplicity is a strength, not a weakness, acknowledge it and claim it as an advantage.

Eliminate Simple Ideas

When trying to solve a problem, you should begin by eliminating simple ideas. There are two reasons why:

- It may be the case that that simple ideas *really* work very well, this happens much more often than you might think.

- Your paper is making the implicit claim "*This is the simplest way to get results this good*". You need to convince the reviewer that this is true, to

Eliminate Simple Ideas: Case Study II

A paper sent to SIGMOD 4 or 5 years ago tackled the problem of **Generating the Most Typical Time Series in a Large Collection**.

The paper used a complex method using wavelets, transition probabilities, multi-resolution properties etc.

The quality of the *most typical time series* was measured by comparing it to every time series in the collection, and the smaller the average distance to everything, the better.

SIGMOD Submission paper algorithm

(a few hundred lines of code, learns model from data)

```
…
X =  DWT(A + somefun(B))
Typical_Time_Series  = X + Z
```

Reviewers algorithm

```
(does not look at the data, and
takes exactly one line of code)

Typical_Time_Series  = zeros(64)
```

Under their metric of success, it is clear to the reviewer (without doing any experiments) that a constant line is the optimal answer for any dataset!

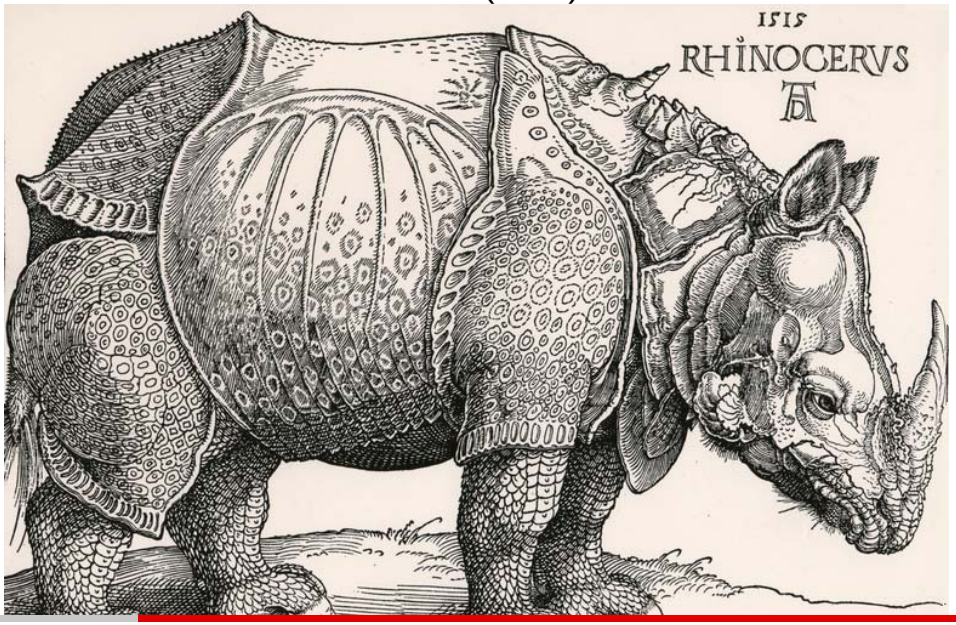
We should always be thinking in the back of our minds, is there a simpler way to do this?

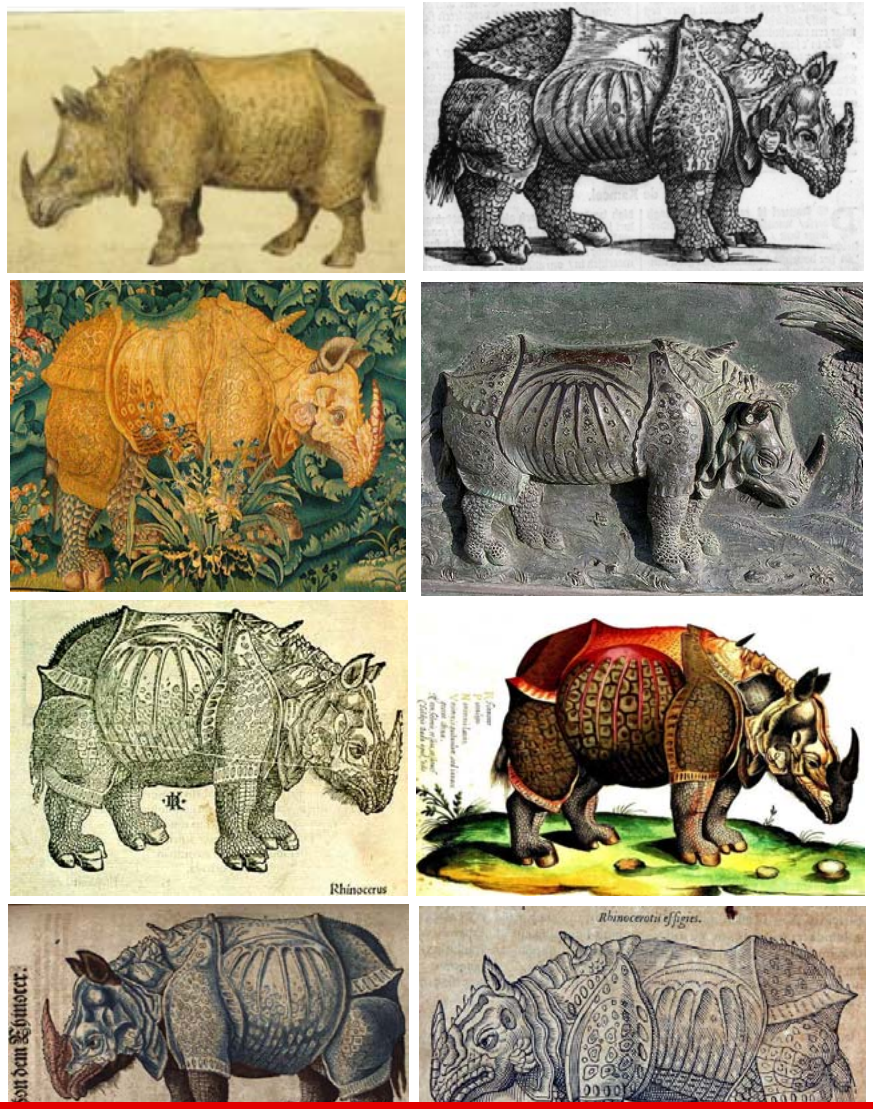
The Importance of being Cynical

In 1515 Albrecht Dürer drew a Rhino from a sketch and written description. The drawing is remarkably accurate, except that there is a spurious horn on the shoulder.

This extra horn appears on every European reproduction of a Rhino for the next 300 years.

Dürer's Rhinoceros (1515)





It Ain't Necessarily So

- Not every statement in the literature is true.
- Implications of this:
  - Research opportunities exist, confirming or refuting "known facts" (or more likely, investigating under what conditions they are true)
  - We must be careful not to assume that it is not worth trying X, since X is "*known*" not to work, or Y is "*known*" to be better than X
- In the next few slides we will see some examples

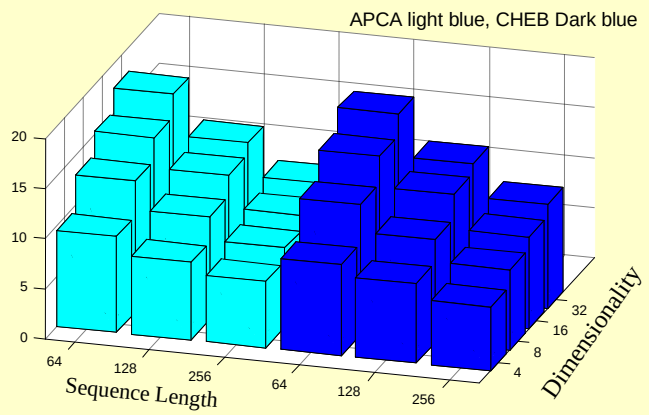
If you would be a real seeker after truth, it is necessary that you doubt, as far as possible, all things.

A SIGMOD Best Paper says..

Our empirical results indicate that Chebyshev approximation can deliver a 3- to 5-fold reduction on the dimensionality of the index space. For instance, it only takes 4 to 6 Chebyshev coefficients to deliver the same pruning power produced by 20 APCA coefficients

Is this really true?

No, actually Chebyshev approximation is slightly worse that other techniques (Ding et al VLDB 08)



The good results were due to a coding bug..

.. Thus it is clear that the C++ version contained a bug. We apologize for any inconvenience caused (note on authors page)

This is a problem, because many researchers have assumed it is true, and used Chebyshev polynomials without even considering other techniques. For example..

(we use Chebyshev polynomial approximation) because it is very accurate, and incurs low storage, which has proven very useful for similarity search. Ni and Ravishankar 07

In most cases, do *not* assume the problem is solved, or that algorithm X is the best, just

A SIGKDD (r-up) Best Paper says..

(my paraphrasing) *You can slide a window across a time series, place all exacted subsequences in a matrix, and then cluster them with K-means. The resulting cluster centers then represent the typical patterns in that time series.*

Is this really true?

No, if you cluster the data as described above *the output is independent of the input* (random number generators are the only algorithms that are supposed to have this property). The first paper to point this out (Keogh et al 2003) met with tremendous resistance at first, but has been since confirmed in dozens of papers.

This is a problem, dozens of people wrote papers on making it faster/better, without realizing it does not work at all! At least two groups published multiple papers on this:

- Exploiting efficient parallelism for mining rules in time series data. Sarker et al 05
- Parallel Algorithms for Mining Association Rules in Time Series Data. Sarker et al 03
- Mining Association Rules from Multi-stream Time Series Data on Multiprocessor Systems. Sarker et al 05
- Efficient Parallelism for Mining Sequential Rules in Time Series. Sarker et al 06
- Parallel Mining of Sequential Rules from Temporal Multi-Stream Time Series Data. Sarker et al 06

In most cases, do *not* assume the problem is solved, or that algorithm X is the best, just

Miscellaneous Examples

Voodoo Correlations in Social Neuroscience. Vul, E, Harris, C, Winkielman, P & Pashler, H.. Perspectives on Psychological Science. Here social neuroscientists criticized for overstating links between brain activity and emotion. This is an wonderful paper.

Why most Published Research Findings are False. J.P. Ioannidis. PLoS Med 2 (2005), p. e124.

Publication Bias: The "File-Drawer Problem" in Scientific Inference. Scargle, J. D. (2000), Journal for Scientific Exploration 14 (1): 91–106

Classifier Technology and the Illusion of Progress. Hand, D. J. Statistical Science 2006, Vol. 21, No. 1, 1-15

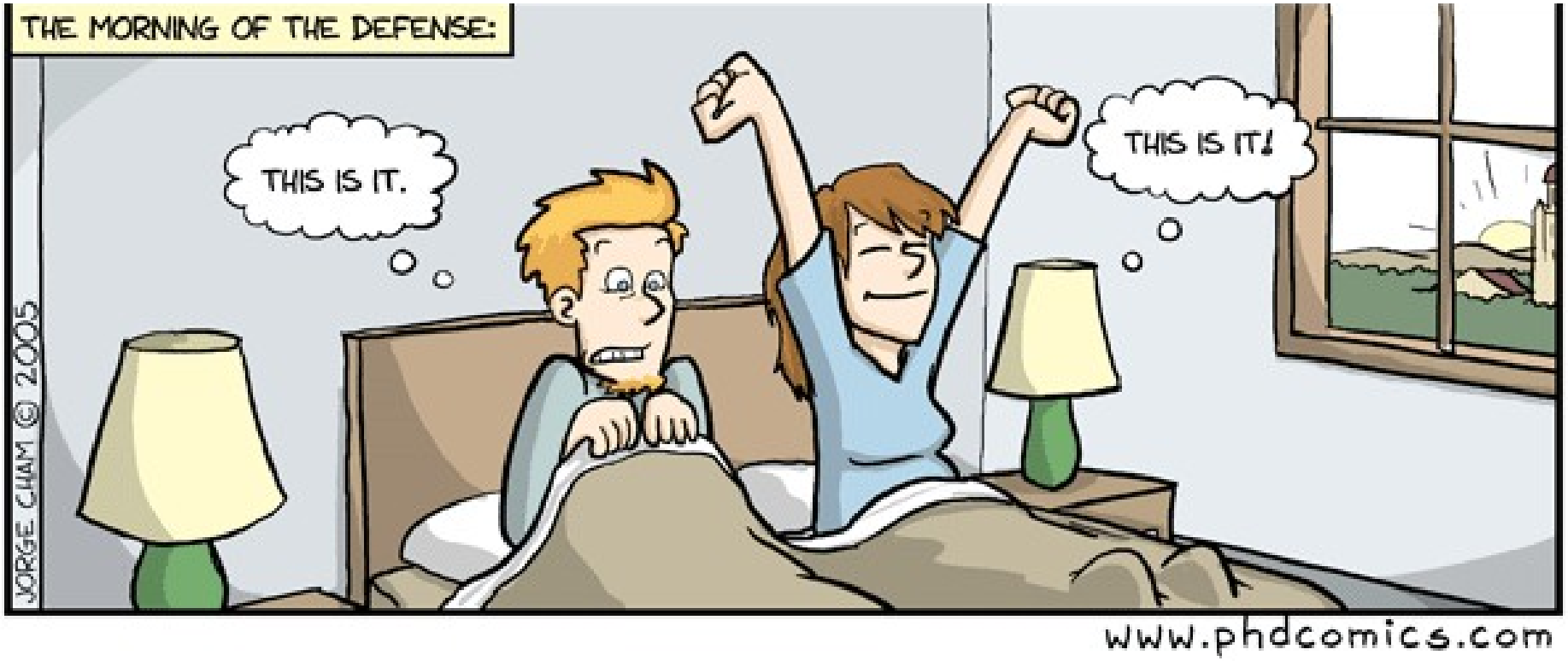
Everything you know about Dynamic Time Warping is Wrong. Ratanamahatana, C. A. and Keogh. E. (2004). TDM 04

Magical thinking in data mining: lessons from CoIL challenge 2000 Charles Elkan

How Many Scientists Fabricate and Falsify Research? A Systematic Review and

CONCLUSION

Conclusion



Good luck!!

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